

Python 環境安裝教學

DSP Lab 2025 autumn Audio and Speech week1

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Outline:

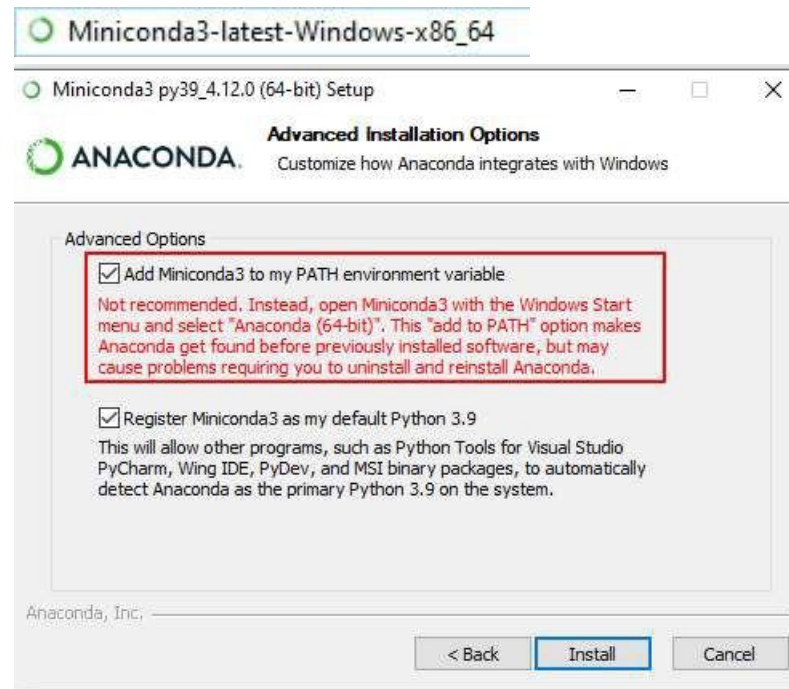
- 相關操作工具
- 安裝說明
- 環境說明
- 工作區設定
- 基礎操作

相關操作工具

- VScode：撰寫程式、連線到WSL或遠端
- Windows Terminal：比較好看方便的命令視窗
- MobaXterm：遠端相關操作(連線、上傳/下載...)
- Conda：環境管理工具

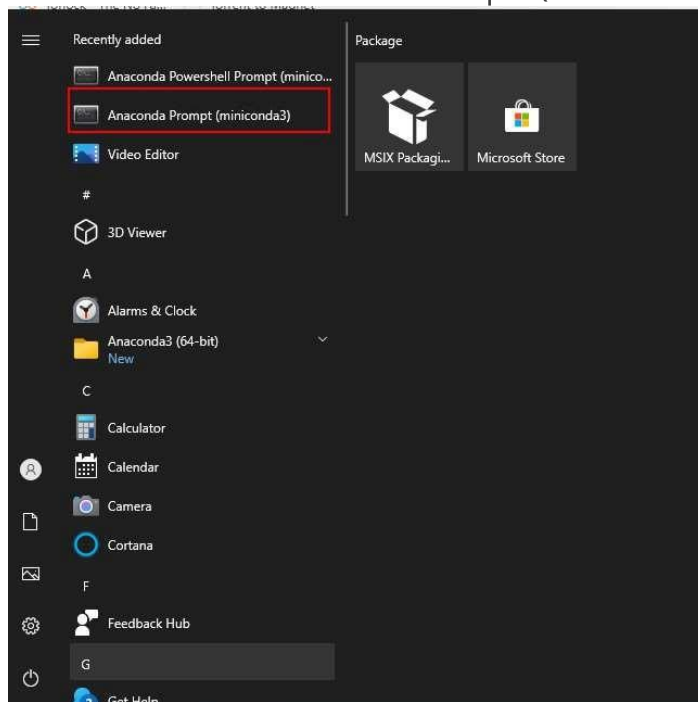
安裝說明(1)

1. 到conda官網下載 (短網址 : url.o-w-o.cc/conda-win)
2. 點開安裝
3. 添加路徑



安裝說明(2)

4. 打開Anaconda Prompt(minicinda3) 輸入指令 conda env list



```
(base) C:\Users\owo>conda env list
# conda environments:
#
base                    * C:\Users\owo\miniconda3
```

環境說明

建立環境

```
1 conda create -n env_name # env_name是環境名稱
2 conda create -n my_env38 python==3.8 # 可以指定版本
3
4 conda activate my_env38 # 切換到對應環境
```

安裝套件 一般主要用pip或conda進行安裝

```
pip install numpy
conda install numpy
```

工作區設定 (1)

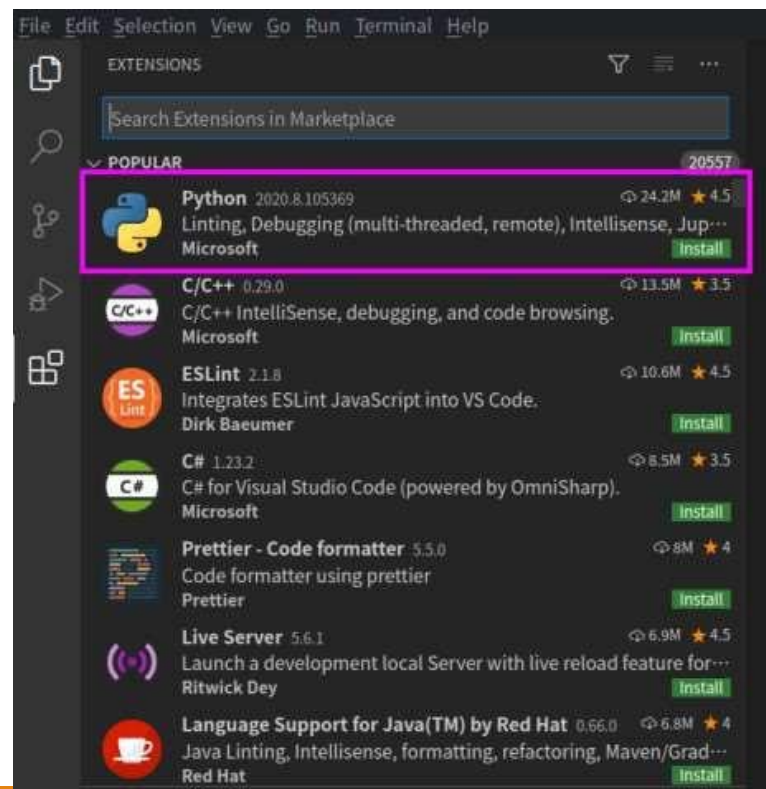
1. 安裝 Visual Studio Code 主程式 <https://code.visualstudio.com/download>
2. 開啟 VS Code，安裝 Python 的 Extension (擴充)

打開左側的擴充欄位，也可以使用 Ctrl + Shift + X 打開



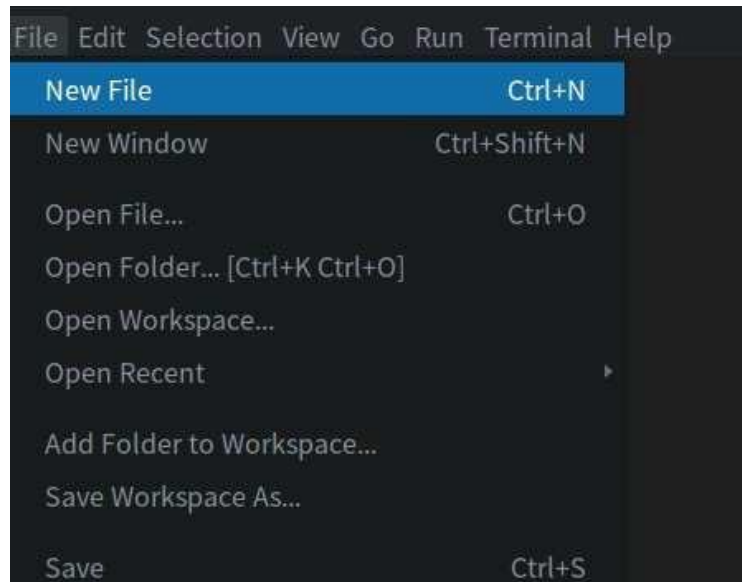
工作區設定 (2)

3. 選擇 Python 擴充並安裝



基礎操作

1. 創建新檔案
2. 將其儲存成 test.py，.py 的副檔名會讓 Python 環境生效。

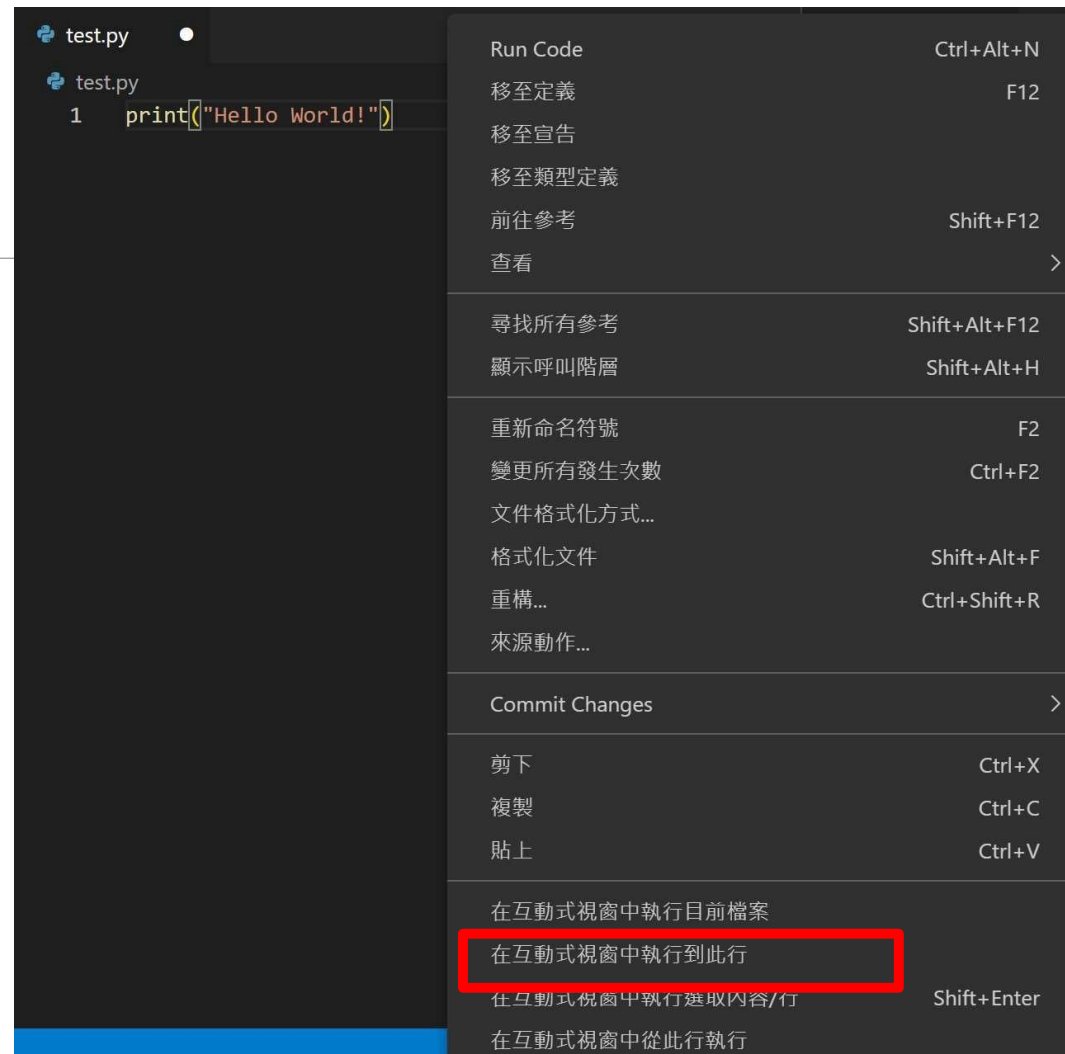


基礎操作

3. `print('Hello World!')` 示範。

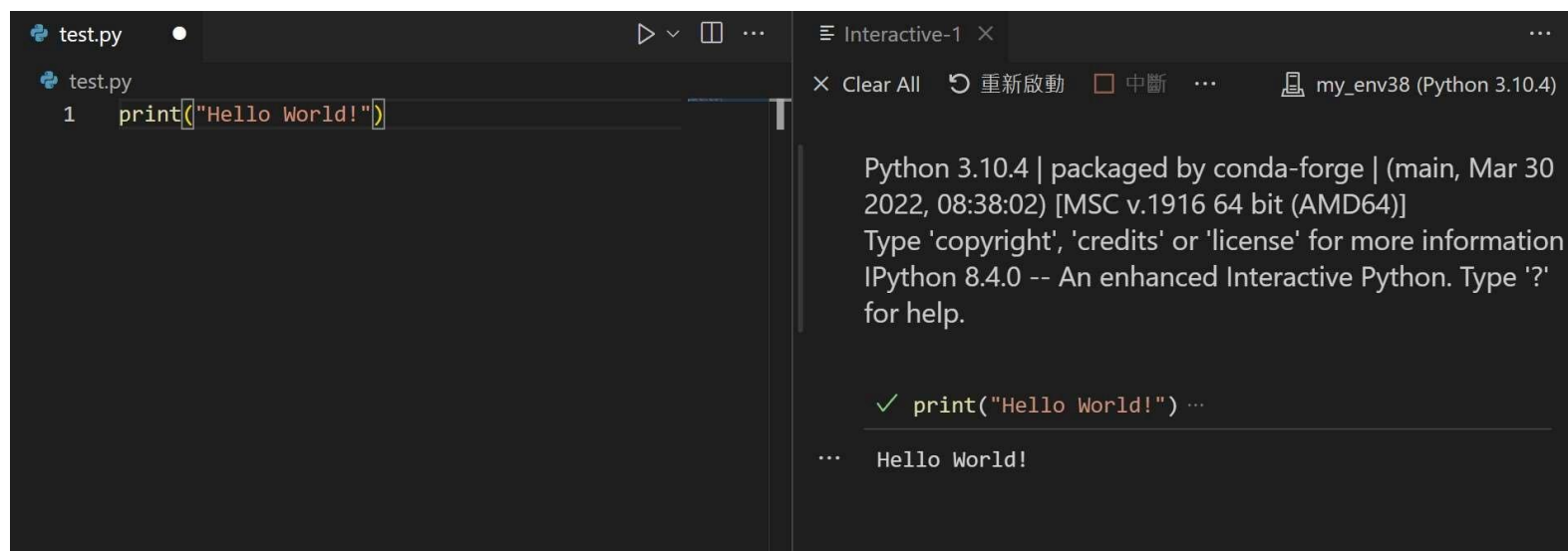
4. 右鍵滑鼠

選擇 Run Python File in Terminal



基礎操作

右邊視窗會跑出結果



The screenshot displays a Jupyter Notebook interface with two main panels. The left panel, titled 'test.py', contains a single line of Python code: `1 print("Hello World!")`. The right panel, titled 'Interactive-1', shows the execution environment 'my_env38 (Python 3.10.4)'. It includes a toolbar with buttons for 'Clear All', '重新啟動' (Restart), '中斷' (Interrupt), and a menu icon. Below the toolbar, the panel displays the IPython version information: 'Python 3.10.4 | packaged by conda-forge | (main, Mar 30 2022, 08:38:02) [MSC v.1916 64 bit (AMD64)]'. It also shows the command 'Type \'copyright\', \'credits\' or \'license\' for more information' and 'IPython 8.4.0 -- An enhanced Interactive Python. Type \'?\' for help.' The execution output is shown as a green checkmark followed by the command `print("Hello World!")` and the result 'Hello World!'.

```
test.py
test.py
1 print("Hello World!")
```

Interactive-1 X ...

X Clear All 重新啟動 中斷 ... my_env38 (Python 3.10.4)

Python 3.10.4 | packaged by conda-forge | (main, Mar 30 2022, 08:38:02) [MSC v.1916 64 bit (AMD64)]
Type 'copyright', 'credits' or 'license' for more information
IPython 8.4.0 -- An enhanced Interactive Python. Type '?' for help.

✓ print("Hello World!") ...

... Hello World!